

C4-2a : Create Database and Tables

Basic SQL - Using DB Browser for SQLite

SQL and SQLite



SQL stands for **S**tructured **Q**uery **L**anguage

The various versions are MySQL, Microsoft Sybase, Oracle, SAP, PostgreSQL

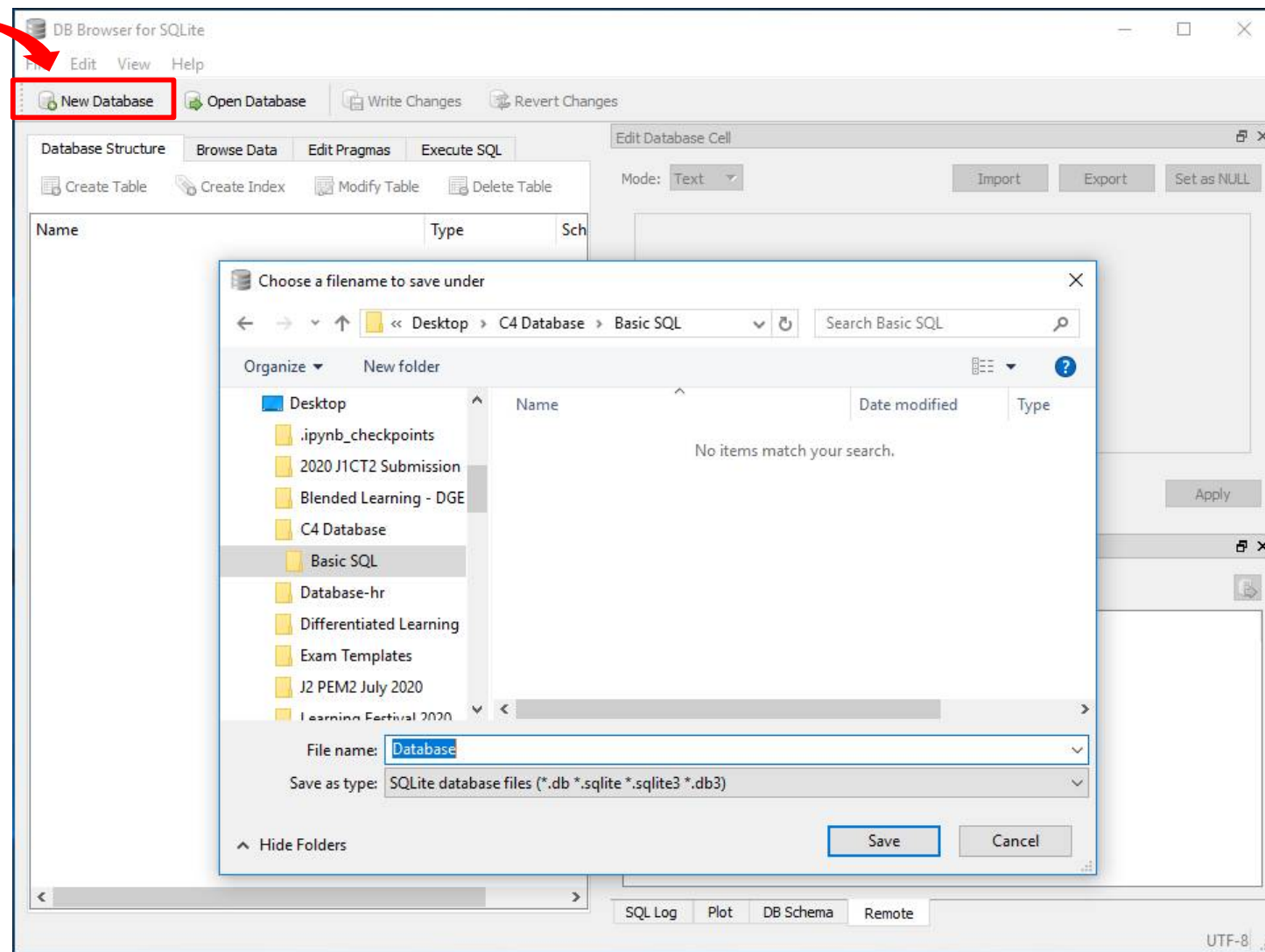
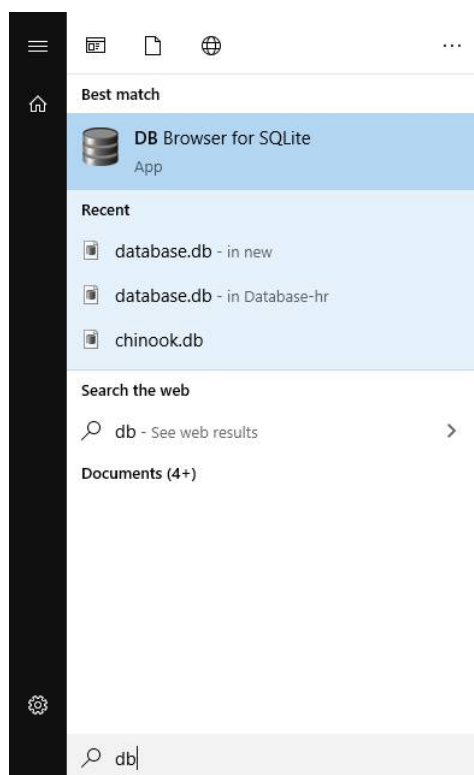
SQL Server is a **client-server database management system** (DBMS) that consists of a database server and client applications that connect to the server.

SQLite is a **serverless DBMS** that can be embedded directly into an application.

All DBMS support the major commands for **C**reate, **R**ead, **U**ppdate and **D**eleete functions (**CRUD**)

DB Browser for SQLite

- Launch DB Browser
- Create a new database and save it as Database.db



Create Table Customers

Table: Customers

Advanced

Fields:

Name	Type	Not	PK	AI	U	Default	Check
Id	INTEGER	☒	☒	☐	☐		
FirstName	TEXT	☒	☐	☐	☐		
LastName	TEXT	☒	☐	☐	☐		
City	TEXT	☐	☐	☐	☐		
Country	TEXT	☒	☐	☐	☐		
Phone	TEXT	☒	☐	☐	☐		

```

1 CREATE TABLE `Customers` (
2   `Id` INTEGER NOT NULL,
3   `FirstName` TEXT NOT NULL,
4   `LastName` TEXT NOT NULL,
5   `City` TEXT,
6   `Country` TEXT NOT NULL,
7   `Phone` TEXT NOT NULL,
8   PRIMARY KEY(`Id`)
9 );
  
```

OK Cancel

Customers
<u>Id</u>
FirstName
LastName
City
Country
Phone

```

CREATE TABLE Customers (
  Id            INTEGER NOT NULL,
  FirstName     TEXT NOT NULL,
  LastName      TEXT NOT NULL,
  City          TEXT,
  Country       TEXT NOT NULL,
  Phone         TEXT NOT NULL,
  PRIMARY KEY(Id)
);
  
```

Create Table Products

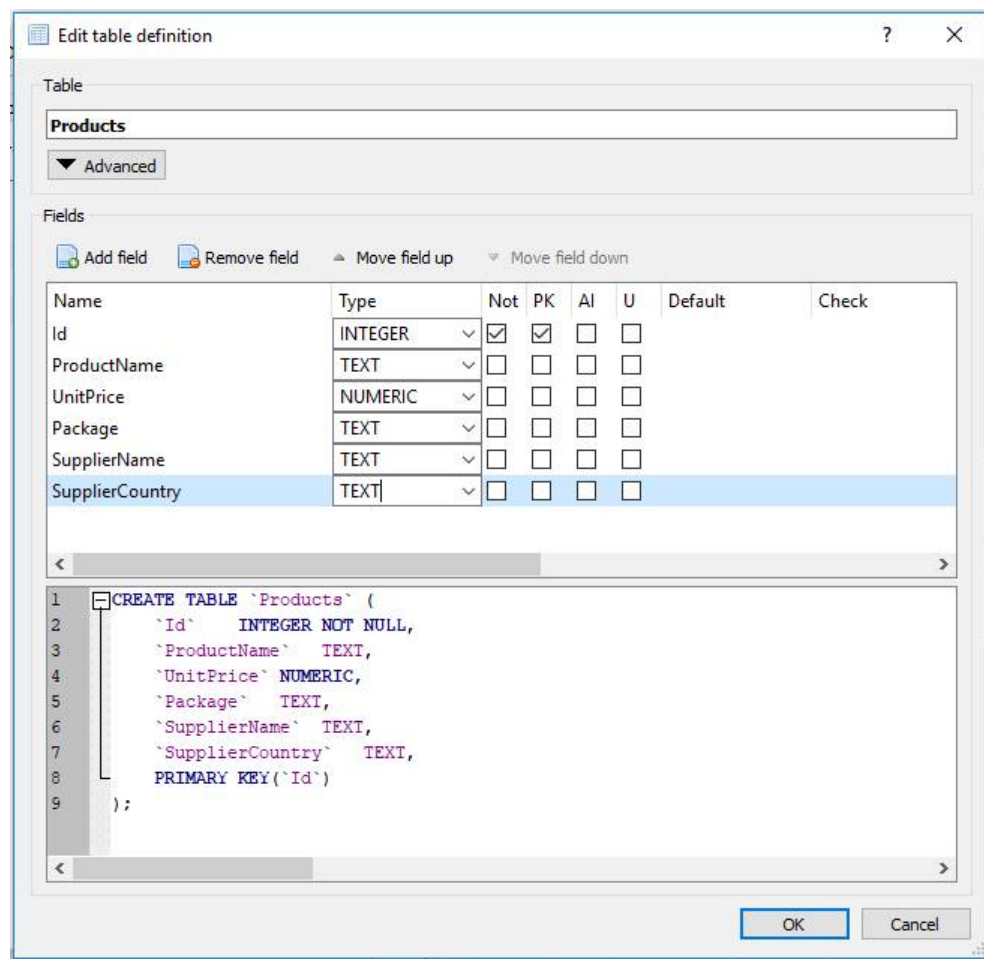


Table: Products

Fields:

Name	Type	Not	PK	AI	U	Default	Check
Id	INTEGER	☒	☒	☐	☐		
ProductName	TEXT	☐	☐	☐	☐		
UnitPrice	NUMERIC	☐	☐	☐	☐		
Package	TEXT	☐	☐	☐	☐		
SupplierName	TEXT	☐	☐	☐	☐		
SupplierCountry	TEXT	☐	☐	☐	☐		

```

1 CREATE TABLE `Products` (
2   `Id` INTEGER NOT NULL,
3   `ProductName` TEXT,
4   `UnitPrice` NUMERIC,
5   `Package` TEXT,
6   `SupplierName` TEXT,
7   `SupplierCountry` TEXT,
8   PRIMARY KEY(`Id`)
9 );

```

Products
<u>Id</u>
ProductName
UnitPrice
Package
SupplierName
SupplierCountry

```

CREATE TABLE Products (
  Id
  ProductName
  UnitPrice
  Package
  SupplierName
  SupplierCountry
  PRIMARY KEY(Id)
);

```

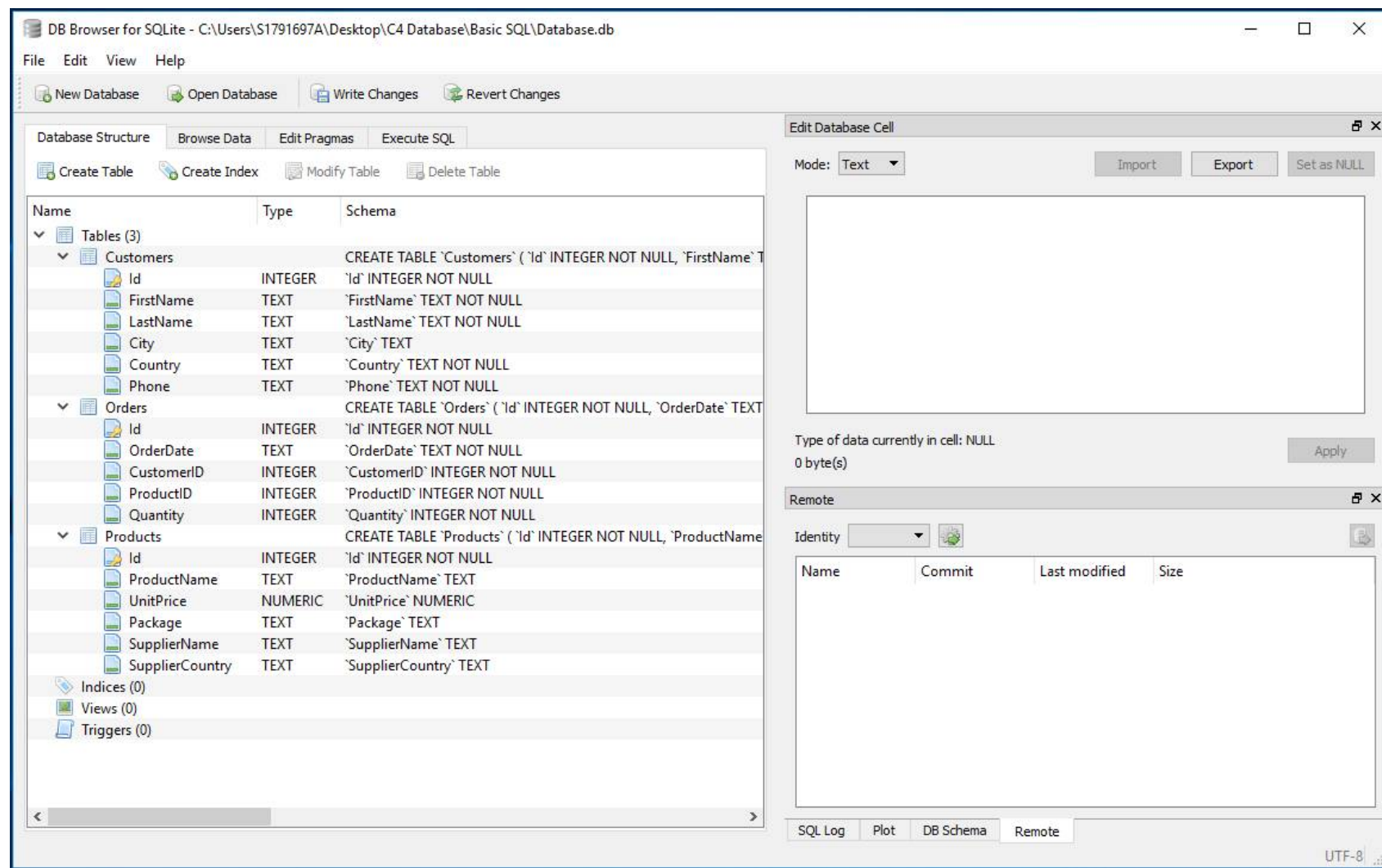
```

INTEGER NOT NULL,
TEXT,
NUMERIC,
TEXT,
TEXT,
TEXT,
TEXT,

```

Database Structure – Table Schema

- Copy the create statements, paste them in a notepad and save it as "Create Tables.sql"

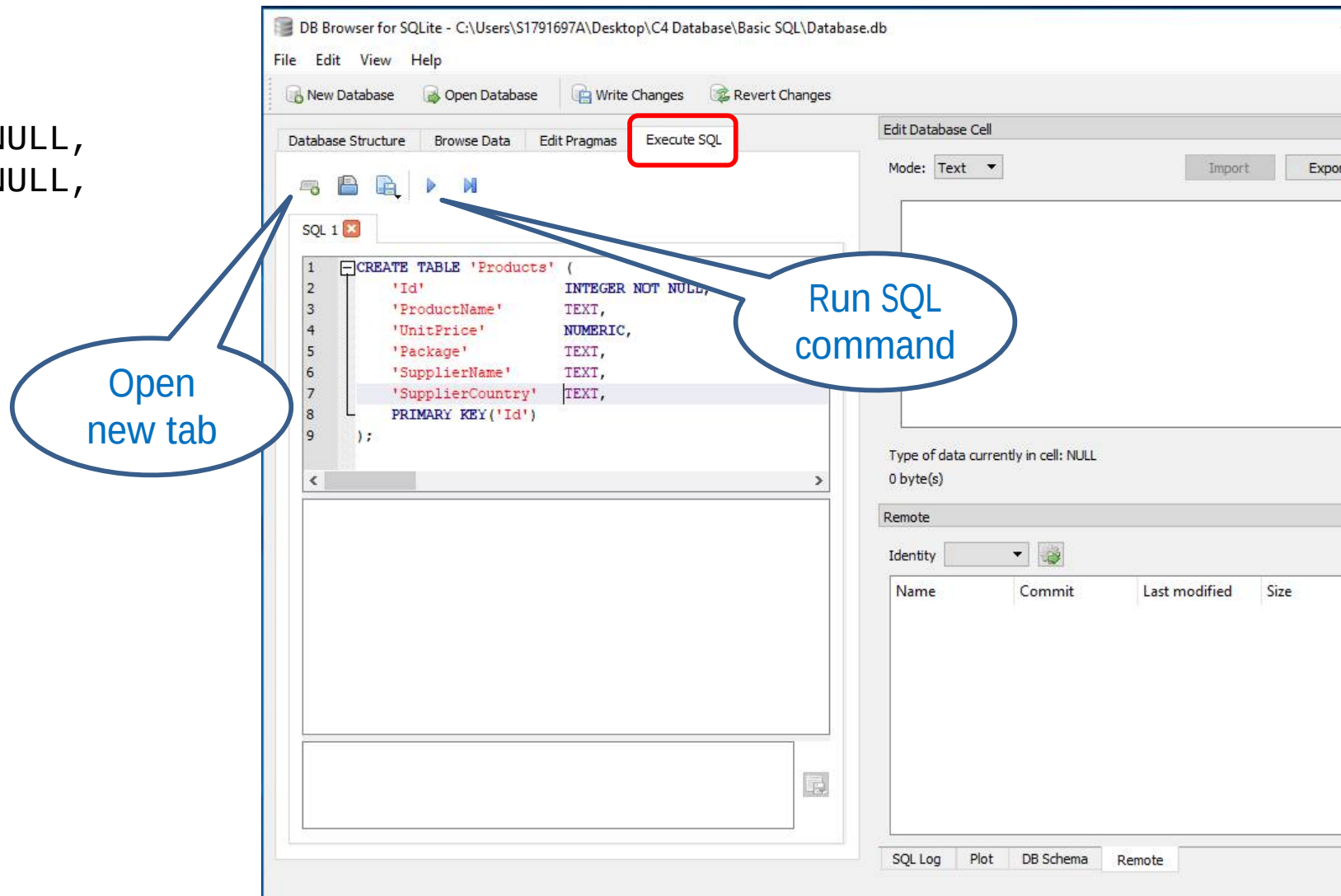


Execute SQL

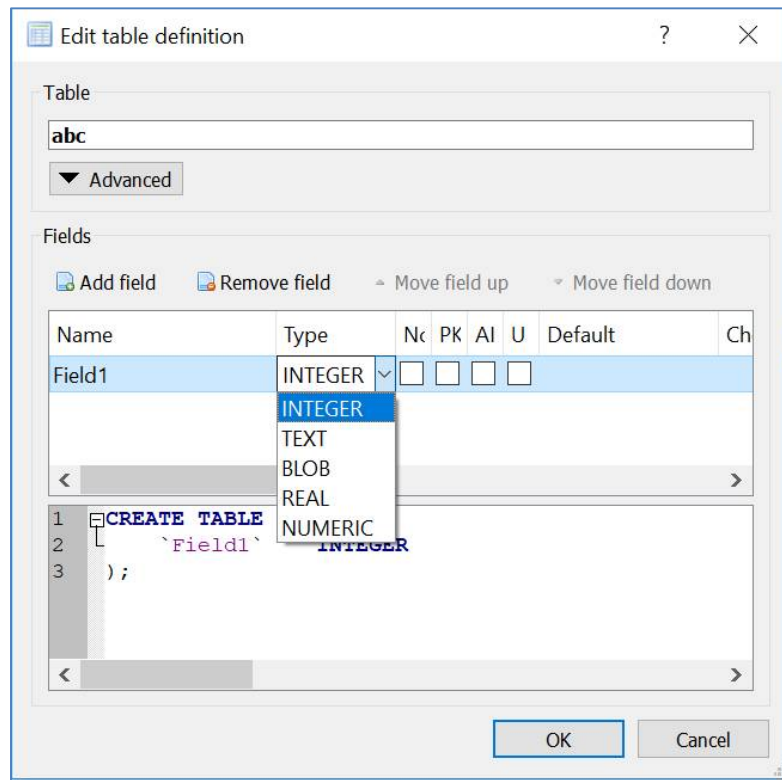
Create Tables using SQL Commands

```
CREATE TABLE Customers (
  Id                INTEGER NOT NULL,
  FirstName          TEXT NOT NULL,
  LastName           TEXT NOT NULL,
  City              TEXT,
  Country            TEXT NOT NULL,
  Phone              TEXT NOT NULL,
  PRIMARY KEY(Id)
);
```

```
CREATE TABLE Products (
  Id                INTEGER NOT NULL,
  ProductName        TEXT,
  UnitPrice          REAL,
  Package            TEXT,
  SupplierName        TEXT,
  SupplierCountry    TEXT,
  PRIMARY KEY(Id)
);
```



Data Types



Data Type	Description
INTEGER	Whole number
TEXT	Text string : CHAR(n), VARCHAR(n)
BLOB	Binary Data Type Eg. Image file
REAL	Decimal (or Floating point) value
NUMERIC	Any other type not listed above. Eg. Boolean, DateTime ...

VARCHAR is variable length, while **CHAR** is fixed length

Boolean values can also be stored as INTEGER, e.g. 0 (False) and 1 (True).

DateTime can also be stored as TEXT, e.g. in the 'YYYY-MM-DD' and 'HH:MM:SS.SSS' formats.

Note: There is no need to go into details about the *Type Affinity* in SQLite (<https://www.sqlite.org/datatype3.html>)